# Topic 4: Acquiring IT/IS Infrastructure Capabilities and Performance Evaluation-Part 2

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## Topic 4: Acquiring IT/IS Infrastructure Capabilities and Performance Evaluation

1. IT/IS Infrastructure
2. Business Value of IS
3. IS Infrastructure and Service Acquiring Process
4. Souring and Outsourcing IS Services and Applications
5. IT/IS Performance Evaluation and Management

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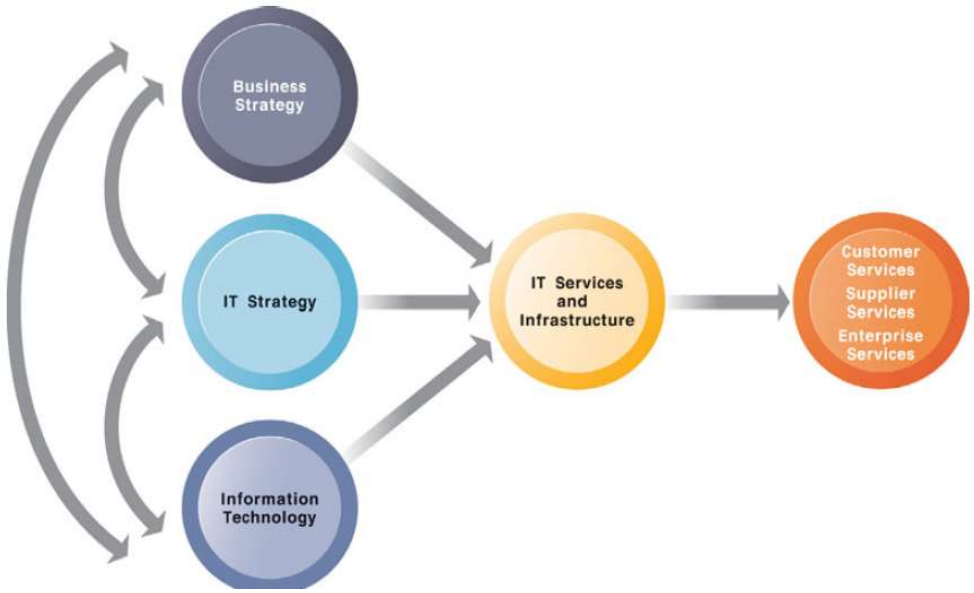
# IT Infrastructure and Business Capability

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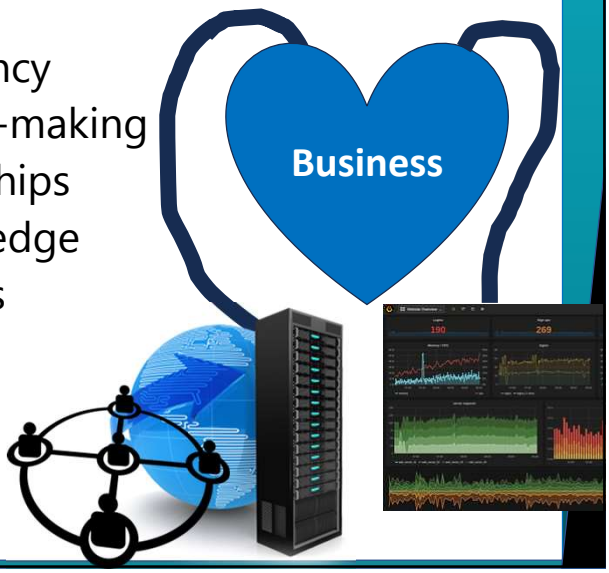
# Business Value of Information Systems

- Enhancing operational efficiency
- Supporting strategic decision-making
- Improving customer relationships
- Gain/maintain a competitive edge
- Streamline business processes
- Managing risks
- Reducing costs
- Fostering collaboration

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## Effective IT Infrastructure

- **Flexible**
  - Connectivity,
  - Compatibility, and
  - Modularity.
- **Robust**
  - Provides data and information to users with the appropriate levels of accuracy, timeliness, reliability, security, and confidentiality
  - Enables employees to perform their duties, having both the available technology and the necessary technological skills.

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## Four Major Factors to be Considered when Acquiring IT Infrastructure

1. Dependability
2. Manageability
3. Adaptability
4. Affordability

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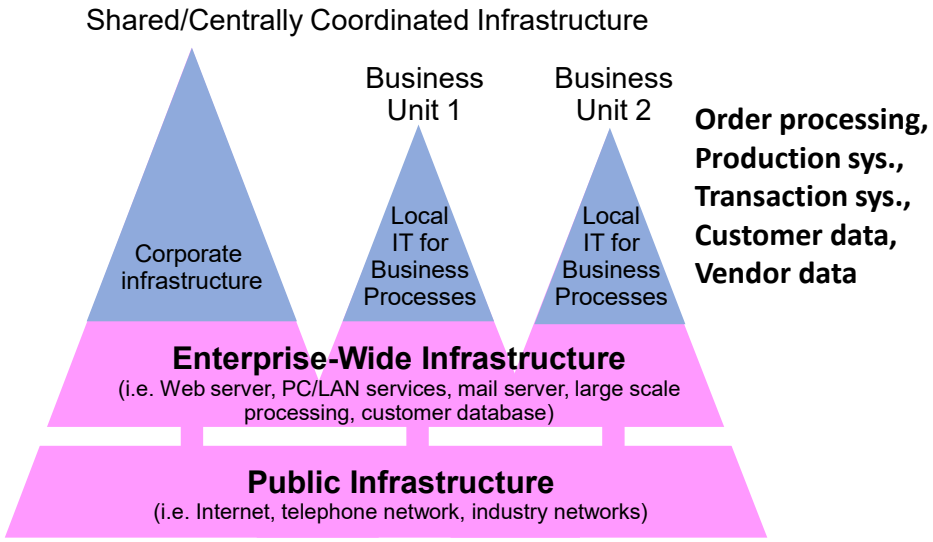
# Levels of IT Infrastructure

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# IT infrastructure Investments

- **Enterprise-wide IT infrastructure**
  - Support long-term , enterprise-wide strategies
  - Responsive to the demands of business-unit strategies
- **Investments at different business units/branch/department**
  - Often made independently of a short-term, catch-up or bleeding-edge strategy
  - Resulting technologies are often incompatible

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# Issues in IT infrastructure Investments

- **Overinvesting**
  - Leads to wasted resources that weigh heavily on the bottom line
- **Underinvesting/ implementing wrong infrastructure**
  - Result in delays
  - Rushed implementations
  - Islands of automation
  - Limited sharing of resources, information and expertise

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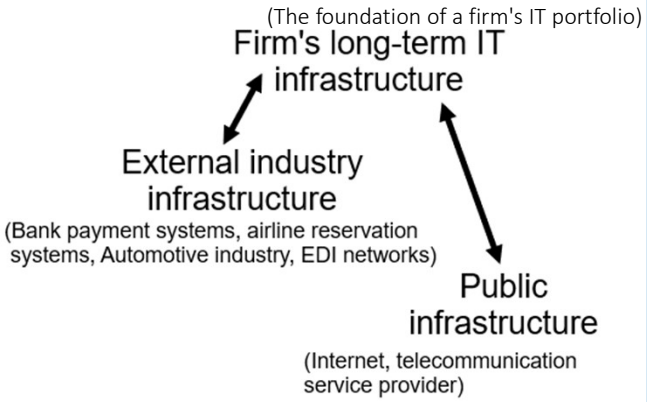
# Take decisions regarding IT infrastructure acquisition considering;

- Strategic issues
- Technical issues
- Competitive issues
- Financial and
- Organizational issues




Consider IT Portfolio and IT architecture

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# IT Portfolio

- A set of managed IT investments allocated to investment strategies,
  - according to an **optimal mix**
  - based on assumptions about future **performance**,
  - balancing the **management goals and risks**.
- All investments in computing and communication technology
- Managed by business and technical management to create business value

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# Components of IT Architecture

| | | |
|---|---|---|
| I T | Business Architecture | Process and activities to meet enterprise's business goals |
| | Application Architecture | How applications are designed and integrated |
| | Data Architecture | How enterprise's data stores are organized and accessed |
| | Technical Architecture | Hardware, software and network to support applications and their interactions |

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# IT Architecture

- Created by the strategic business/IT planning process
- A conceptual design/blueprint describing the information-processing assets needed to meet business objectives
- Includes rules that govern IT assets and the information associated with them
- Evolves over time- not set in concrete and must be reviewed
- Usually, a technical guideline rather than a tool for decision making
- Has to cope with business uncertainty and technological changes

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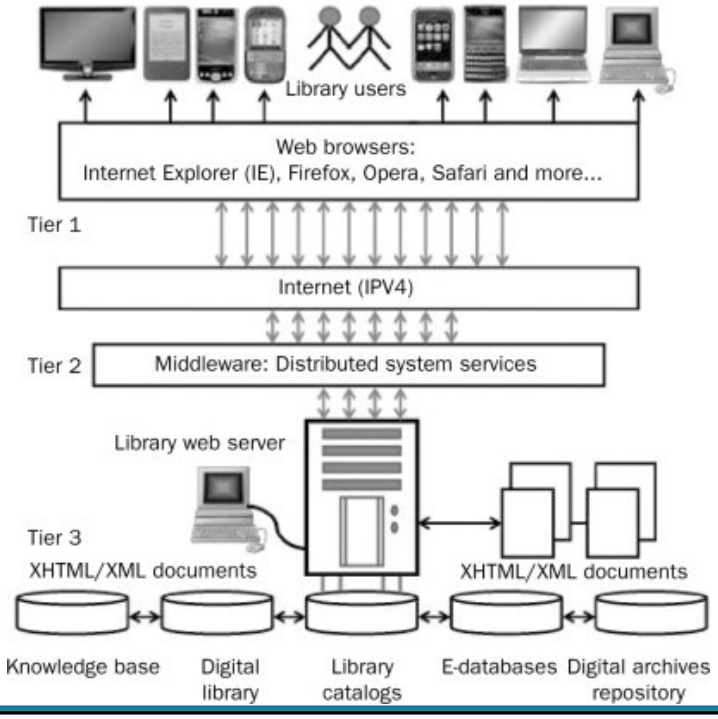
# Example: Three-tier Library Technology Architecture

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## Uses and Benefits of IT Architecture

- Maintain a close alignment between IT deliverables and business requirements.
- Improve ability to respond quickly to business changes.
- Develop closer partnerships between business and IT groups.
- Reduce the risk of failed or unnecessary information systems.
- Reduce the complexity of existing ISs.
- Improve the efficiency of new IT systems.
- Ensure that legal and regulatory requirements are being met.

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## HOMEWORK: ACTIVTY -1

1. Read the following webpage and prepare a note

https://news.microsoft.com/europe/2020/05/05/microsoft-announces-a-1-billion-digital-transformation-plan-for-poland-including-access-to-local-cloud-services-with-first-datacenter-region/

2. Describe the importance of converged systems and hyper-converged systems

Reference: https://www.nutanix.com/info/converged-vs-hyperconverged-infrastructure

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# 4.3 Acquiring IS resources and capabilities

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## IS service/application capabilities

- **IS application capability** - The abstract perspective on the **functional behaviour** required to support the business, i.e. what the IS application **functionality** will be to support the Business Processes.

- **IS service capability** refers to the **effectiveness and efficiency,** i.e. User expectation-satisfaction and whether it can do what is promised.

# Sourcing IS services and applications

- It is the **process of finding, evaluating and engaging suppliers** of services or applications
- **Effectiveness of IS sourcing** will impact IS application/service capability, especially in system management and maintenance

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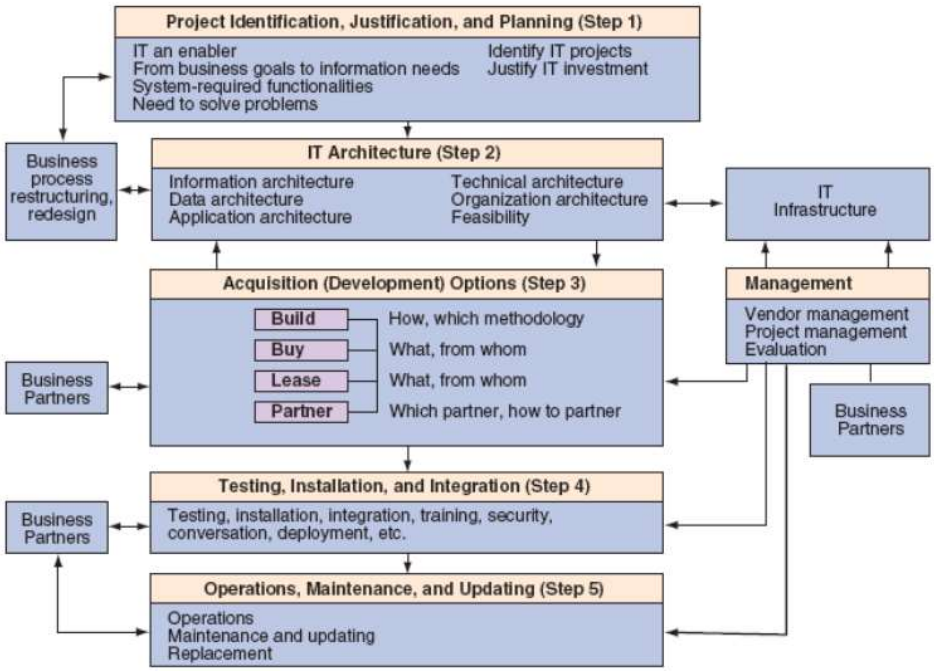
# Steps in IS application acquisition

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## Acquisition options/strategies

- **Build** the system in-house or have a vendor build a custom-made system
- **Buy** an existing application and install it
- **Lease** standard software- Use the s/w and pay a rental to the leasing company
- Enter into a **partnership or alliance** that will enable the company to use someone else's application
- Use a combination of these approaches

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## ACTIVITY

- Define the following terms;
  - Insourcing
  - Outsourcing
  - Co-sourcing
  - Offshoring

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# In-house development (in-sourcing)

**Types:** Bespoke development and End-user development

**Development approaches**
- Build from scratch
- Build from components
- Integrating applications

**Development methods**
- Systems development life cycle (SDLC)
- Prototyping methodology
- Agile Software Development methods





# Advantages of build option

- Best fit with the company requirements
- Have control over software improvements
- Have all of the required features
- Maintain core competencies and a level of quality service
- Make a distinction with other companies

## Disadvantages of build option

- Required more IT personnel
- High overhead cost
- Time consuming
- Problem with usability of the system
- High switching cost
- Difficult to update to newer technology





## End-user development

- Enables end users to design or customize the user interface and functionality of IS
- Development and use of ISs by people outside the IS department.
- Beneficial to both workers and the organization as a whole.
- Quality and cost may be at risk unless proper controls are installed.

## Buying (Off-the-Shelf Solution)

**Factors to be considered;**

- Vendor stability
- System upgrades
- Customer support provided by vendor
- Hardware and software requirements
- Required customization of the base software





## Advantages of buying

- Many different types of off-the-shelf software are available.
- Much time can be saved by buying rather than building.
- The company can know what it is getting before it invests in the software.
- The company is not the first and only user.
- The vendor updates the software frequently.
- The price is usually much lower for a buy option.

## Disadvantages of buying

- Software may not exactly meet the company's needs.
- Software may be difficult or impossible to modify, or it may require huge business process changes to implement.
- The company will not have control over software improvements and new versions.
- Purchased software might be difficult to integrate with existing systems.
- Vendors may drop a product or go out of business.



## Buying: Software License

- Get a license to use the software
- One-time payment – have lifetime use
- Get the ownership - be an asset of the organization
- Version locking -may not get updates
- Self-hosted –get on premise easy access
- Perpetual – Licenses does not get expired
- Vendor independent
- Offline access – No need to be online to use the software

## Buying: Software Subscription

- Lower Upfront Cost
- Recurring payments – can pay as you use the software
- Get temporary access – Can use as long as subscription is active
- Up-to-date versions – Get access to the latest versions
- Online support
- Scalability – Can upgrade or downgrade the subscription plan depending on the current requirements

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## Outsourcing

- Part or all of the IS activities are performed by one or more external suppliers (Looff, 1998) **based on a contractual agreement** (Muhic andWintzler, 2011)
- Outsourcing, can allow companies to do business more efficiently and effectively

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## Factors behind Outsourcing IS

- Cost
- Inadequate internal expertise and talent pool
- Rapid technology change
- Poor chargeback system for IS services
- Emphasis on core competency
- Top management discomfort with technology
- Management innovation
- Management imitation
- Changing goals and objectives





## Outsourcing process

1. Strategic intent
2. Level of outsourcing
3. Vendor selection
4. Contract Negotiation
5. Project Management and monitoring
6. Project renewal or termination

## Advantages of Outsourcing

- Cost Reduction (financial benefits)
- Access to world class specialist providers (technological benefits)
- Improved focus on core business (operational benefits)
- Subcontracting of workload
- Better risk management



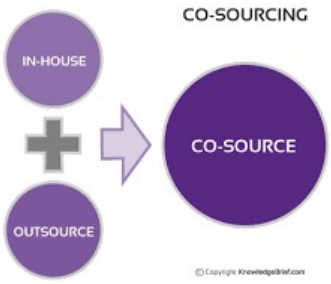


## Disadvantages of Outsourcing

- Loss of organizational competencies
- Reduction in quality of services
- Cost escalation from unforeseen expenses

# Co- sourcing

- A partnership between a firm and an outside organization, a professional service provider
- Acquire specialized services while keeping the functional control in-house
- Does not replace the existing staff. Professional service firm's job is over when the project is finished
  - A way for a company to maintain control while cutting costs and maximizing internal audit capabilities.

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# Offshoring

- Acquire services or resources from another country
- Can produce goods or services in a cheaper country
- Can take advantage of certain tax or tariff relief in some countries
- Retain full control and ownership

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## Factors to be considered in making the software acquisition decision

- Suitability to the company's business needs
- Scalability: can handle increased users, customers, or requests
- Flexibility: Adaptability and modifiability
- Customer support: Installation and training support and other after sales services
- Interoperability: Does data interchange work?
- Time-to-implementation : off-the-shelf s/w is the quickest
- Implementation risks: installation or update issues
- Property rights: Ownership
- Cost of ownership: Purchase price + costs of operation

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## Review Exercises

1. List the major IS acquisition strategies and briefly describe each.
2. Compare buy option against lease option.
3. List the in-house development approaches and briefly describe each.
4. Describe end-user development and list its advantages and limitations.
5. Describe the benefits of co-sourcing.
6. What are the four fundamental business decisions that an organization should make when acquiring IS?

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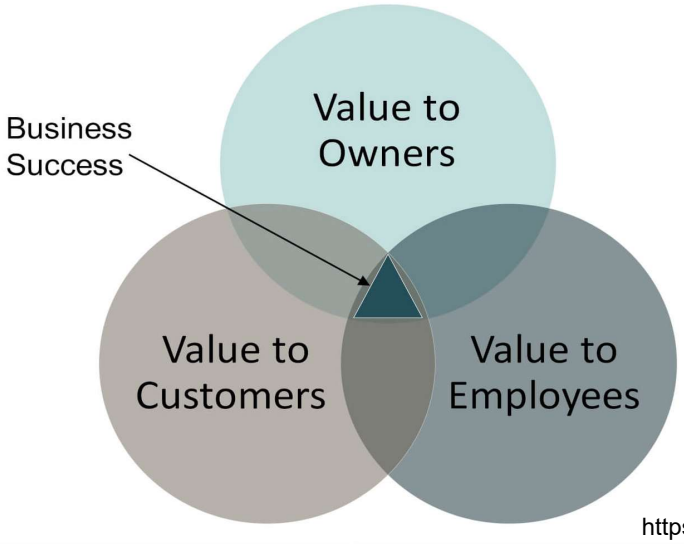
# System Performance Evaluation

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https://cfoperspective.com

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# Factors Used to Assess IS Performance

- Usability: Efficiency of IT services and the overall quality of a consumer's experience
- Functionality(validity): Ability of a product to perform the work it's intended to do.
- Reliability: Performs without interruption or errors
- Performance: Effectiveness of the work. Measured by **accuracy and speed**
- Security: Protection of sensitive information from unauthorized access, attacks and damage.
- Agility: Ability to respond to outside incentives such as pressure to change or adapt and the ability to create and respond to new opportunities.

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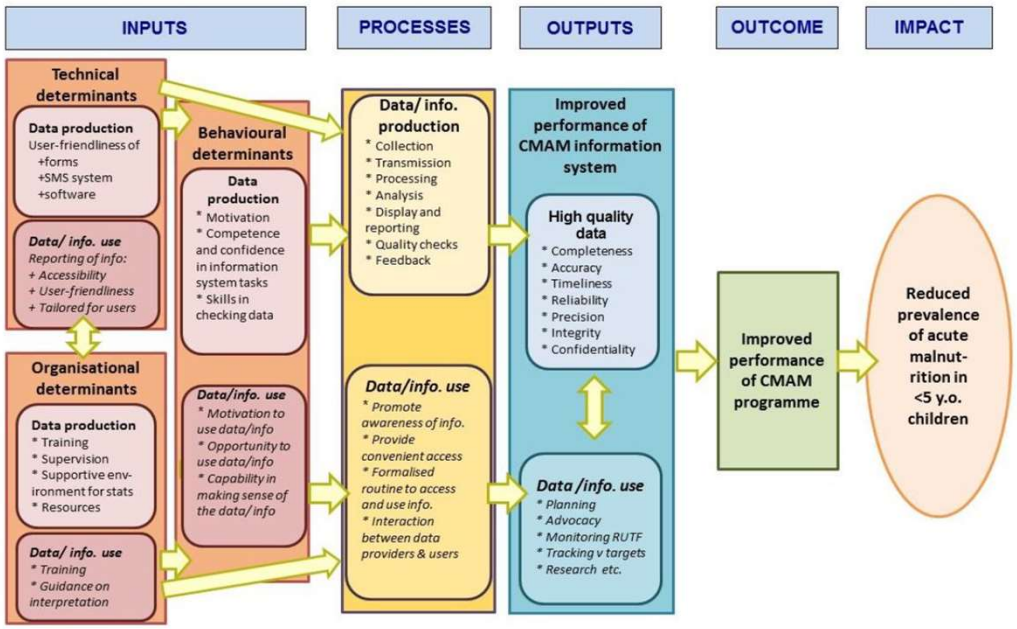
## e.g. Framework for the Assessment of an Information System

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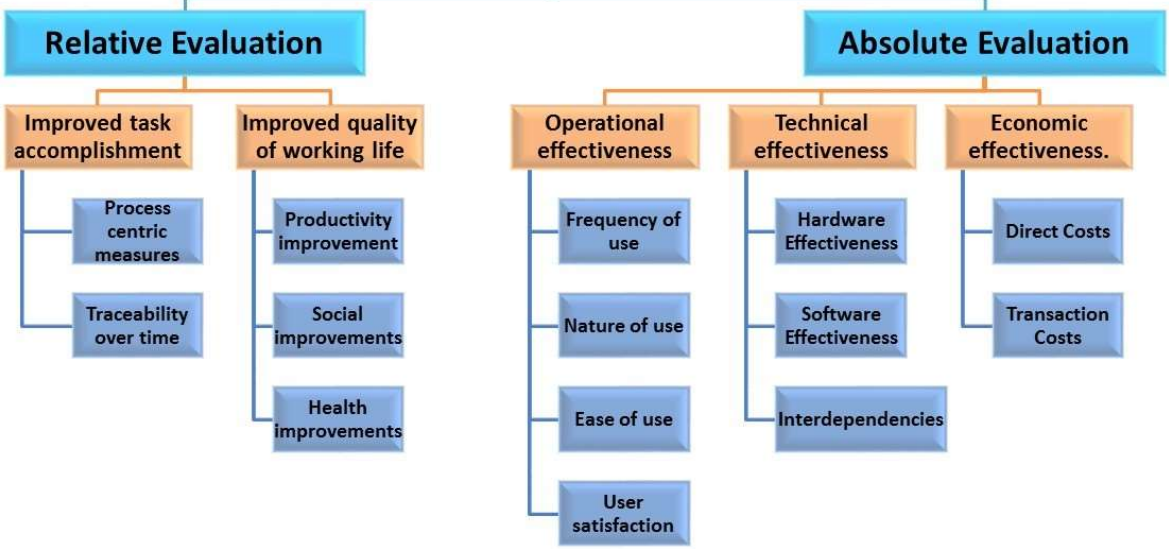
## Information System Effectiveness

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# Activity

- See the IS evaluation form in the following paper and prepare the list of IS quality factors considered in **ISO25010**

https://www.stepacademic.net/ijcsr/article/download/100/52/